CLASSIFICATION OF BIPARTITE OBSERVABLES III: SIMPLICITY, MIXTURES AND QUANTUM ERROR CORRECTION

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Abstract. The paradigm of a simple unstable theory in the sense of Shelah is the theory of the random graph. This graph, also called the Rado graph, is the Fraïssé limit of the finite graphs: the countable graph, unique up to isomorphism, that contains every finite graph as an induced subgraph and in which every isomorphism between finite induced subgraphs extends to an automorphism. The stabilizer formalism of quantum error correction, the protection of quantum information against noise, rests on the Pauli group. On countably many qubits, or two-level quantum systems, this group consists of the Pauli strings, the tensor products of Pauli matrices on finitely many of the qubits, multiplied by phases. We show that, taken modulo phases and with anticommutation as its relation, the Pauli group is the Fraïssé limit of the finite vector spaces over the two-element field with an alternating form, and that its theory, like that of the random graph, is simple and unstable. Anticommutation computes the syndromes of stabilizer codes: a stabilizer code detects an error by measuring checks, commuting Pauli strings that generate its stabilizer group, and the syndrome of the error records which checks anticommute with it. We show that three notions of model theory have counterparts in these codes. Clifford circuits, the unitary operations that permute the Pauli strings under conjugation, up to phases, implement the homogeneity of the Pauli group: every isomorphism between finite-dimensional substructures extends to an automorphism induced by such a circuit, and every stabilizer code has an encoding circuit of this kind. Forking recognizes the logical errors, those that have zero syndrome and yet alter the encoded information: an error with zero syndrome is a logical error exactly when its type over the stabilizer group and a set of logical operators, which generate the errors of zero syndrome modulo the stabilizer group, forks over the stabilizer group. The independence theorem of Kim and Pillay becomes the gluing of syndromes: if two sets of checks are independent over a common set, then any two syndromes, each realized by some error on its own set of checks, that agree on the common set are realized by a single error. A bipartite observable is a bounded self-adjoint operator on the tensor product of the Hilbert spaces of two parties, Alice and Bob; with each preparation of Alice and each probe of Bob, it yields a real number, the response. The Rado graph and the Pauli group define bipartite observables, which we call the Rado observable and the Pauli observable. We prove that, once mixed preparations of Alice are admitted, the real-valued theory of each of the two observables has the tree property of the second kind, a combinatorial configuration incompatible with simplicity. We show that the two observables differ in the preparations that witness this property, and that Bob's probes account for the difference. The probes of the Pauli observable are the Pauli channels, the probability laws of random Pauli errors. For finitely many linearly independent checks, the probability that a Pauli channel produces an error with a given nonzero syndrome is the difference of the responses of two uniform mixtures of checks, and actual preparations, as opposed to the idealized preparations of an elementary extension, witness the tree property of the second kind. The responses of the Rado observable depend on a probe only through its marginals, the probabilities that it assigns to adjacency with the individual vertices. For this observable, every witness of the tree property of the second kind for a quantifier-free formula contains idealized preparations.

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1. Introduction

The paradigm of a simple unstable theory is the theory of the random graph [She80, GIL02]. A first-order theory is simple, in the sense of Shelah, when no formula has the tree property, a branching pattern of consistent and inconsistent conditions that we describe below; the theory of the random graph is unstable because its edge relation has the independence property: for any finitely many vertices and any choice among them, some vertex is adjacent to the chosen ones and to none of the others. The random graph has two modes of presentation: as the Rado graph, it is the Fraïssé limit of the finite graphs, that is, the countable graph, unique up to isomorphism, that contains every finite graph as an induced subgraph and in which every isomorphism between finite induced subgraphs extends to an automorphism [Fra54, Rad64]; as a random object, it is the graph to which a random network of couplings on countably many sites, in which each pair of sites is coupled independently with probability one half, is isomorphic with probability one [ER63]. Quantum error correction, the protection of quantum information against noise, rests on a structure of its own. In a stabilizer code, errors are detected by measuring checks, which are Pauli strings: tensor products of factors on the individual qubits, or two-level quantum systems, each factor being the identity, a bit flip, a phase flip or their product. The checks of a code commute, and the code stores quantum information in their common eigenspace for the eigenvalue 1. It suffices to correct errors that are themselves Pauli strings, since a code that corrects a set of errors corrects every linear combination of them (see [Got97]).

To make this precise, let $\mathbb{F}_2$ be the two-element field and let n be a positive integer. The Hilbert space of a qubit is $\mathbb{C}^2$, on which the bit flip X and the phase flip Z, two of the Pauli matrices, exchange the two vectors of the standard basis and change the sign of the second, respectively. A Pauli string on n qubits is a tensor product of n factors $X^{x_k}Z^{z_k}$, for $k = 1, \dots, n$, with $x_k, z_k \in \mathbb{F}_2$, multiplied by a phase; up to its phase, it is determined by its label, the vector $(x, z) \in \mathbb{F}_2^{2n}$ with $x = (x_1, \dots, x_n)$ and $z = (z_1, \dots, z_n)$. Let ω be the form on $\mathbb{F}_2^{2n}$ given by $\omega((x, z), (x', z')) = x \cdot z' + z \cdot x'$ for $x, z, x', z' \in \mathbb{F}_2^n$, where $x \cdot z' = \sum_{k=1}^{n} x_k z'_k$; it is bilinear, alternating and nondegenerate, and it is the standard symplectic form on $\mathbb{F}_2^{2n}$. Two Pauli strings on n qubits either commute or anticommute, and their labels decide which: the strings anticommute exactly when ω takes the value 1 at their labels. We disregard phases

and identify each Pauli string with its label. Measuring a check g on a state that g fixes, after an error e has acted on the state, reveals whether the two strings anticommute: the syndrome bit of e with respect to g is 1 if they anticommute and 0 if they commute, and in terms of labels it is $\omega(g, e)$. The syndrome of e with respect to finitely many checks is the vector of its syndrome bits.

On countably many qubits, the labels of the Pauli strings that act nontrivially on only finitely many qubits form a vector space over $\mathbb{F}_2$ of countably infinite dimension, on which the form ω, given by the same formula, is nondegenerate. This countable symplectic space is, under our identification, the Pauli group on countably many qubits taken modulo phases, which we call simply the Pauli group. In Section 3 we show that, with this form, the Pauli group is the Fraïssé limit of the finite alternating spaces over $\mathbb{F}_2$, the finite vector spaces over $\mathbb{F}_2$ with an alternating bilinear form, and that its theory, like that of the random graph, is simple and unstable. For each of the two structures, we prove simplicity by exhibiting an explicit relation of independence between two sets over a third and verifying for it the axioms of Kim and Pillay, which ensure that the theory is simple (see Fact 2.2). The Pauli group is thus the quantum counterpart of the random graph, and in this paper we study the two paradigms side by side.

In Section 4 we give a dictionary between model theory and quantum error correction, and we prove the three results on which it rests. The checks of a stabilizer code generate its stabilizer group; an error is undetectable when it commutes with every check, and an undetectable error is a logical error when it alters the encoded information, that is, when it lies outside the stabilizer group. Clifford unitaries, the unitary operators that map Pauli strings to Pauli strings under conjugation, up to phases, implement the homogeneity of the Pauli group: every isomorphism between finite-dimensional substructures of the Pauli group extends to an automorphism induced by a Clifford unitary on finitely many qubits, and every stabilizer code has an encoding circuit that is a Clifford unitary. The type of an error over a set of Pauli strings records which elements of the set anticommute with it and how, if at all, it is a product of them, and forking is the model-theoretic notion of an extension of a type that adds information. Forking recognizes the logical errors: an undetectable error is a logical error exactly when its type over the stabilizer group and a set of logical operators, which generate the undetectable errors modulo the stabilizer group, forks over the stabilizer group. The independence theorem of Kim and Pillay becomes a rule for syndromes: if two sets of checks are independent over a common set, then any two syndromes, each realized by some error on its own set of checks, that agree on the common set are realized by a single error. As first-order structures the two paradigms are alike: both are Fraïssé limits, and both have simple unstable theories. We show that the bipartite observables that they define tell them apart.

This paper is the third of a series on bipartite observables. A bipartite observable is a bounded self-adjoint operator on the tensor product of the Hilbert spaces of two systems, one prepared by Alice and the other probed by Bob. A preparation of Alice, a state of her system, averages the observable to an operator on Bob's space, and a probe of Bob, a state of the algebra generated by these operators, assigns to the operator of each preparation a real number, the response of the preparation at the probe. In the first paper [DI26a] we studied the stability of an observable in the sense of Shelah, which requires that the expectation of the observable, for independently prepared systems, not depend on which of the two systems is idealized first, that is, replaced by a limit of states given by density operators along an ultrafilter. In the second [DI26b] we showed that an observable is stable exactly when its idealized responses, the responses of its idealized preparations, are continuous functions

on the space of probes, and that it is NIP, which means that it lacks the independence property, exactly when they are of the first Baire class, that is, pointwise limits of sequences of continuous functions. The idealized responses of a NIP observable form a Rosenthal compactum, a compact space of functions of the first Baire class in the topology of pointwise convergence, and the classification of these compacta due to Todorčević applies to them. Simplicity is another dividing line of Shelah, one of the properties by which he classified first-order theories; it neither implies nor follows from NIP, and a theory is stable exactly when it is both simple and NIP. It is, moreover, a property of a different kind: stability and NIP of an observable are characterized by properties of the closure of its family of responses in the topology of pointwise convergence, whereas simplicity concerns the joint consistency of conditions on an unknown element, such as a probe. A theory fails to be simple exactly when, for some formula and some integer $k \geq 2$, there is a tree of instances of the formula, indexed by the finite sequences of natural numbers, whose branches are consistent while any k children of each node are jointly inconsistent. In error correction, the unknown element is an error, and the conditions are statements about its syndrome.

In Section 5 we define the observables of the two paradigms. Both are diagonal observables: the Hilbert spaces of Alice and Bob have orthonormal bases indexed by countable sets of labels, and the observable reports whether a relation holds between Alice's label and Bob's. In the Rado observable, Alice and Bob each hold a site of the network, that is, a vertex of the Rado graph, and the observable reports whether the two sites are coupled; in the Pauli observable, Alice holds a check and Bob an error, and the observable reports whether the two anticommute. A mixed preparation of Alice, a density operator on her space, enters a diagonal observable only through its diagonal, a probability measure on her labels, and the point masses at single labels are her pure preparations; a single mixed preparation can therefore spread its weight over arbitrarily many labels, and its response is affine in the mixture. Bob's probes differ in the two observables, as we establish in Lemma 5.5 and Proposition 5.4: for the Rado observable they are the Radon probability measures on the space of all sets of neighbors, that is, the random sets of neighbors, and for the Pauli observable they are the Pauli channels, the probability laws of random Pauli errors of arbitrary support. Every set of vertices is thus a pure probe of the Rado observable, and we say that its pure responses, the responses of pure preparations, are freely independent. The pure probes of the Pauli observable, its idealized errors, are the Pauli strings of arbitrary support, and the syndrome sector of a syndrome of finitely many checks is the set of idealized errors with that syndrome. The depolarizing noise of rate p, for $0 < p \leq 1$, which acts independently on each qubit, leaving it unchanged with probability $1 - p$ and applying a bit flip, a phase flip or their product, each with probability $p/3$, is a Pauli channel concentrated on the Pauli strings of infinite support, which are not actual errors; it models the noise of a quantum memory on countably many qubits.

The real-valued theory of an observable is the theory of the structure formed by its mixed preparations, its probes and its response, in the real-valued logic of Keisler [Kei23], whose formulas take real values. A formula has the tree property of the second kind when, for some integer $k \geq 2$, there is an array of instances of the formula in which any k instances of a row are jointly inconsistent while every path, which chooses one instance from each row, is consistent. In real-valued logic these conditions are inequalities with a gap: a sequence of instances is uniformly k-inconsistent when, for some reals $\alpha < \beta$, each instance takes a value at least β somewhere while no k of them exceed α at a common point. In Section 6 we prove that the real-valued theory of every diagonal observable with the independence property, and in particular of the Rado observable and of the Pauli observable, has the tree property of

the second kind: the difference between the responses to the uniform mixtures on a half of a finite set of labels and on the complementary half reaches its maximum, 1, at a probe exactly when the probe answers 1 on the first half and 0 on the second, and the differences given by two halves that meet in a quarter of the set are never both greater than $1/2$. Actual preparations are those of the structure itself, and idealized preparations are the new elements of its elementary extensions. In Sections 7 and 8 we show that, for the Pauli observable, the tree property of the second kind is witnessed by an array of actual preparations, while, for the Rado observable, every array that witnesses it for a quantifier-free formula contains idealized preparations. In Section 9 we compare mixing in our sense with the affine logic of Ben Yaacov, Ibarlucía and Tsankov, in which a mixture of structures is a direct integral, and Section 10 contains open questions.

Theorem 1.1 (Main results).

(A) *The Rado graph and the countable symplectic space over $\mathbb{F}_2$, which is the Pauli group on countably many qubits modulo phases, are simple and unstable (Theorems 3.1 and 3.3).*

(B) *Let a stabilizer code be given, let S be the set of labels of its stabilizer group, and let L be a set of logical operators of the code. An undetectable error is a logical error exactly when its type over $S \cup L$ forks over S; all logical errors have the same type over S, and this type does not fork over the empty set (Theorem 4.5 and Proposition 4.7). Every isomorphism between finite-dimensional substructures of the Pauli group extends to an automorphism induced by a Clifford unitary on finitely many qubits, and every stabilizer code has an encoding circuit that is a Clifford unitary (Theorem 4.3 and Corollary 4.4). The independence theorem becomes the gluing of syndromes: if two sets of checks are independent over a common set, then any two syndromes, each realized by some error on its own set of checks, that agree on the common set are realized by a single error (Proposition 4.8). The probes of the Pauli observable are the Pauli channels, and the depolarizing noise of every positive rate on countably many qubits is concentrated on the Pauli strings of infinite support, which are idealized errors and not actual ones (Proposition 5.4).*

(C) *For every diagonal observable whose pure responses, the responses of single labels, have the independence property, and in particular for the Rado observable and the Pauli observable, the real-valued theory of the observable has the tree property of the second kind, for the formula that is the difference of two responses. Since the independence property of a diagonal observable passes to its pure responses, it follows that a simple diagonal observable is NIP (Theorem 6.1).*

(D) *For the Pauli observable, the indicator of the syndrome sector of every nonzero syndrome of finitely many linearly independent checks is, on pure probes, the difference of the responses of two uniform mixtures of checks, and the tree property of the second kind is witnessed by actual preparations: for blocks of linearly independent checks, the conditions that a given check is the first of its block to fire, that is, to anticommute with the unknown error, form the rows of a witnessing array; in the bit-flip repetition code, whose checks compare neighboring qubits of a chain, the first check to fire marks the first domain wall of a bit flip, the first boundary between flipped and unflipped qubits (Proposition 7.1, Theorem 7.2 and Example 7.4).*

(E) *For the Rado observable, and more generally for every diagonal observable with freely independent pure responses, such as the observable of the random bipartite graph, the Fraïssé limit of the finite bipartite graphs, the conditions on a probe given by instances of a quantifier-free formula with actual mixtures as parameters are compatible along every sequence: no such sequence is uniformly k-inconsistent, for any integer $k \geq 2$. Hence, for these observables, every array that witnesses the tree property of the second kind of the formula in (C) contains idealized preparations (Theorems 8.1 and 8.3, and Corollary 8.2).*

The difference between (D) and (E) lies in the probes. For the Pauli observable we show in Proposition 5.4 that the pure probes are the Pauli strings of arbitrary support, which form a compact group. Alice's pure responses are then, up to an affine change, the characters of this group. Products of characters are characters, and the indicators of syndrome sectors can therefore be computed by averaging: for finitely many linearly independent checks and

a nonzero syndrome, the indicator of its syndrome sector is, on pure probes, the average response over those elements of the group generated by the checks which anticommute with the errors of the sector, minus the average response over those which commute with them (see Proposition 7.1); this identity is the counterpart, for responses, of the familiar expression of the projector onto a syndrome sector as an average over the stabilizer group [Got97], although it requires the checks only to be linearly independent, not to commute. With it, a single mixture of checks can single out the first check of a block to fire, that is, to anticommute with a given error: for linearly independent checks $g_0, g_1, \dots$ and each $j \in \mathbb{N}$, the uniform mixture on the coset of g_j modulo the span of the earlier checks has response 1 at an error exactly when g_j is the first of these checks to anticommute with the error. For the Rado observable, by contrast, we show in Lemma 5.5 that the pure probes are all the sets of neighbors, which form the whole Cantor cube, the space of all subsets of the vertex set with the product topology, and that the pure responses are its coordinate functions. Averages of coordinates depend on a probe only through its marginals, the probabilities with which it declares the individual vertices neighbors; in the proof of Theorem 8.1, a gliding hump argument, which passes to a subsequence whose weights lie, up to small errors, on a common finite set and on pairwise disjoint finite blocks, shows that no sequence of conditions given by bounded affine combinations of responses of actual mixtures is uniformly k-inconsistent, for any $k \geq 2$. In the paradigm, the Rado graph, every array of mixtures that witnesses the tree property of the second kind for a quantifier-free formula therefore contains idealized preparations; in its quantum counterpart, the Pauli group, the linear relations among the labels, which make the syndrome bit of a product of checks the sum of their syndrome bits, allow actual preparations to witness the same property.

2. Simple theories and the theorem of Kim and Pillay

We begin by fixing the vocabulary. We work in a monster model $\mathbb{M}$ of a complete first-order theory T; sets of parameters are small.

Definition 2.1. Let $\varphi(x, y)$ be a formula and let an integer $k \geq 2$ be given.

(1) A sequence $(b_j)_{j\in\mathbb{N}}$ is *k-inconsistent for φ* if for all $j_1 < \cdots < j_k$ there is no x with $\varphi(x, b_{j_1}) \wedge \cdots \wedge \varphi(x, b_{j_k})$.

(2) The formula φ has the *tree property* if, for some $k \geq 2$, there is a tree $(b_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ such that for every $\eta \in \mathbb{N}^{\mathbb{N}}$ the set $\{\varphi(x, b_{\eta\restriction n}) : n \in \mathbb{N}\}$ is consistent and for every s the sequence $(b_{s^\frown j})_{j\in\mathbb{N}}$ is k-inconsistent for φ.

(3) The formula φ has the *tree property of the first kind* if there is a tree $(b_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ such that for every $\eta \in \mathbb{N}^{\mathbb{N}}$ the set $\{\varphi(x, b_{\eta\restriction n}) : n \in \mathbb{N}\}$ is consistent and for all incomparable $s, t \in \mathbb{N}^{<\mathbb{N}}$ the set $\{\varphi(x, b_s), \varphi(x, b_t)\}$ is inconsistent.

(4) The formula φ has the *tree property of the second kind* (TP_2) if, for some $k \geq 2$, there is an array $(b_{ij})_{i,j\in\mathbb{N}}$ such that every row $(b_{ij})_{j\in\mathbb{N}}$ is k-inconsistent for φ and for every $f\colon \mathbb{N} \to \mathbb{N}$ the set $\{\varphi(x, b_{if(i)}) : i \in \mathbb{N}\}$ is consistent.

(5) The theory T is *simple* if no formula has the tree property.

Every stable theory is simple, and the tree property of the second kind implies the tree property; by a theorem of Shelah, a theory is simple exactly when it has neither TP_2 nor the tree property of the first kind (see [She90, Theorem III.7.11]). A simple theory is stable if and only if it is NIP (see [She80, GIL02]), so a simple unstable theory has the independence property. To prove simplicity we do not build trees; instead, we exhibit an independence relation directly.

Fact 2.2 (Kim and Pillay). *Let $\downarrow$ be a ternary relation on small subsets of $\mathbb{M}$, written $A \downarrow_C B$, and assume that it satisfies the following:*

(i) invariance *under automorphisms of $\mathbb{M}$;*
(ii) symmetry*: the relation $A \downarrow_C B$ holds if and only if $B \downarrow_C A$ holds;*
(iii) monotonicity and transitivity*: the relation $A \downarrow_C BD$ holds if and only if both $A \downarrow_C B$ and $A \downarrow_{CB} D$ hold;*
(iv) finite character*: the relation $A \downarrow_C B$ holds if and only if $a \downarrow_C b$ holds for all finite tuples a from A and b from B;*
(v) local character*: for every finite tuple a and every set B there is $B_0 \subseteq B$ with $|B_0| \leq |T|$ and $a \downarrow_{B_0} B$;*
(vi) extension*: for every tuple a and all sets $C \subseteq B$ there is a' with $\mathrm{tp}(a'/C) = \mathrm{tp}(a/C)$ and $a' \downarrow_C B$;*
(vii) independence theorem over models*: if M is a model, $B \downarrow_M D$, $a_1 \downarrow_M B$, $a_2 \downarrow_M D$ and $\mathrm{tp}(a_1/M) = \mathrm{tp}(a_2/M)$, then there is a with $\mathrm{tp}(a/MB) = \mathrm{tp}(a_1/MB)$, $\mathrm{tp}(a/MD) = \mathrm{tp}(a_2/MD)$ and $a \downarrow_M BD$.*

Then the theory T is simple and the relation $\downarrow$ coincides with nonforking independence (see [KP97, Theorem 4.2]*).*

3. The two paradigms

Here we prove Theorem 1.1(A): the Rado graph and the Pauli group on countably many qubits, taken modulo phases and with commutation as its relation, are simple unstable structures. In the subsections below we show that both are Fraïssé limits with quantifier elimination, and that in both the relation of Kim and Pillay is read off from the closure operator: equality for the graph, linear span for the Pauli group.

3.1. The Rado graph.

The class of finite graphs has the amalgamation property, since two graphs that share an induced subgraph can be glued along it without new edges. Its Fraïssé limit is the Rado graph (Γ, E), and a countable graph is this limit if and only if it has the *extension property*: for all finite disjoint $\Gamma_1, \Gamma_0 \subseteq \Gamma$ there are infinitely many $v \in \Gamma \setminus (\Gamma_1 \cup \Gamma_0)$ adjacent to every vertex of Γ_1 and to no vertex of Γ_0 [Fra54, Rad64]. The theory of (Γ, E) is $\aleph_0$-categorical and eliminates quantifiers, so the type of a tuple over a set D is determined by its equalities with elements of D and its edges to D and within itself. By saturation, the monster model has the extension property for small disjoint sets Γ_1, Γ_0.

The relation $E(x, y)$ has the independence property: for distinct $a_1, \ldots, a_n$ and every set of indices among $1, \ldots, n$, the extension property gives a vertex adjacent to a_i exactly when i is in the set. Hence the theory of the Rado graph is unstable.

Theorem 3.1. *The theory of the Rado graph is simple, and the relation $A \downarrow_C B$ holds if and only if $A \cap B \subseteq C$.*

Proof. We verify the hypotheses of Fact 2.2. The first five are set algebra, and extension and the independence theorem follow from the extension property of $\mathbb{M}$. Invariance, symmetry and finite character are immediate. Monotonicity and transitivity hold together, since the conditions $A \cap B \subseteq C$ and $A \cap D \subseteq C \cup B$ say jointly that every element of $A \cap (B \cup D)$ lies in C. For local character, given a finite tuple a and a set B, the finite set $B_0 = B \cap a$ satisfies $a \cap B \subseteq B_0$, which is the relation $a \downarrow_{B_0} B$.

For extension, let a and $C \subseteq B$ be given, and write $a = (c, a_0)$, where c lists the entries of a in C. The type of a_0 over C prescribes only the equalities and edges of a_0 within itself and to C. Hence the extension property of $\mathbb{M}$, applied one entry at a time with the excluded set

enlarged by $B \setminus C$, gives a tuple a_0' disjoint from B with the same type over C, and the tuple $a' = (c, a_0')$ satisfies $a' \cap B \subseteq C$.

For the independence theorem, let M, B, D, a_1 and a_2 be as in Fact 2.2(vii). Since $\mathrm{tp}(a_1/M) = \mathrm{tp}(a_2/M)$, the two tuples have the same entries in M and the same remaining equalities; write $a_\ell = (m, a_\ell')$ with a_ℓ' disjoint from M. The hypotheses $a_1 \mathop{\smile\!\!\!\!\!\!|}_M B$ and $a_2 \mathop{\smile\!\!\!\!\!\!|}_M D$ say that a_1' is disjoint from B and a_2' from D. We look for a tuple x, disjoint from $M \cup B \cup D$, whose edges within itself and to M are those of a_1' (equivalently, of a_2'), whose edges to $B \setminus M$ are those of a_1', and whose edges to $D \setminus M$ are those of a_2'. These prescriptions concern disjoint sets of vertices, because $B \cap D \subseteq M$; the extension property of $\mathbb{M}$ therefore provides such an x, and by quantifier elimination the tuple $a = (m, x)$ is as required. □

3.2. **Pauli strings.** A qubit is the Hilbert space $\mathbb{C}^2$, and its Pauli matrices are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad XZ = -ZX.$$

A Pauli string on n qubits is, up to a phase, a tensor product $X^{x_1}Z^{z_1} \otimes \cdots \otimes X^{x_n}Z^{z_n}$ with $x_k, z_k \in \mathbb{F}_2$, and it is labeled by the vector $(x, z) \in \mathbb{F}_2^{2n}$; products of strings correspond to sums of labels. For a label $v = (x, z)$ we write P_v for the Hermitian string $\mathrm{i}^{x \cdot z} X^{x_1}Z^{z_1} \otimes \cdots \otimes X^{x_n}Z^{z_n}$, where $x \cdot z$ is the number of indices k with $x_k = z_k = 1$; it satisfies $P_v^2 = I$, and the label $(1, 1)$ of one qubit gives $P_{(1,1)} = \mathrm{i}XZ = Y$. Two strings either commute or anticommute, and

$$P_{(x,z)}P_{(x',z')} = (-1)^{\omega((x,z),(x',z'))} P_{(x',z')}P_{(x,z)}, \qquad \omega((x, z), (x', z')) = x \cdot z' + z \cdot x'.$$

The form ω is bilinear, alternating and nondegenerate on $\mathbb{F}_2^{2n}$. The *syndrome* of an error e with respect to checks $g_1, \ldots, g_m$ is the vector $(\omega(g_1, e), \ldots, \omega(g_m, e))$: measuring the checks reveals which of them anticommute with the error. On countably many qubits we let $V = \bigoplus_{k \in \mathbb{N}} \mathbb{F}_2^2$, with the same form; its elements are the Pauli strings that act nontrivially on finitely many qubits.

3.3. **The Pauli group as a Fraïssé limit.** In the language with $+$, 0 and a binary relation for $\omega(x, y) = 1$, substructures are $\mathbb{F}_2$-subspaces with the restricted form. Finite alternating spaces amalgamate: if $A \subseteq B$ and $A \subseteq C$, write $B = A \oplus B'$ and $C = A \oplus C'$, and declare $\omega(B', C') = 0$ on $A \oplus B' \oplus C'$. Every finite alternating space embeds in a nondegenerate one, and by the theorem of Witt [Art57] every isometry between subspaces of a finite nondegenerate alternating space extends to the whole space. A back-and-forth argument along symplectic bases therefore shows that (V, ω) is the Fraïssé limit of the finite alternating spaces over $\mathbb{F}_2$. Its theory is $\aleph_0$-categorical and eliminates quantifiers, so the type of a tuple a over a set D is determined by the linear relations of a over $\mathrm{span}(D)$ and the values of ω on $\mathrm{span}(aD)$. The algebraic closure of a set is its span.

Lemma 3.2. *If $a_1, \ldots, a_n \in V$ are linearly independent and $\varepsilon_1, \ldots, \varepsilon_n \in \mathbb{F}_2$, then there is $b \in V$ with $\omega(a_i, b) = \varepsilon_i$ for $i = 1, \ldots, n$.*

Proof. The vectors a_i lie in a finite-dimensional subspace W spanned by part of the symplectic basis, and the form is nondegenerate on W. Hence the functionals $\omega(a_i, \cdot)$ restricted to W are linearly independent, and the linear map $b \mapsto (\omega(a_i, b))_{i \le n}$ from W to $\mathbb{F}_2^n$ is onto. □

By Lemma 3.2, the formula $\omega(x, y) = 1$ has the independence property: any set of syndromes is realized by some error. Hence the theory of (V, ω) is unstable. By saturation, the monster model $\mathbb{M}$ realizes, over each small subspace S, every finite-dimensional alternating space $S \oplus \mathrm{span}(x_1, \ldots, x_n)$; in particular, for every linear functional f on S there is b with $\omega(\cdot, b) = f$ on S.

Theorem 3.3. *The theory of (V,ω) is simple, and the relation $A \mathop{\smile\!\!\!\!\!\!|}_C B$ holds if and only if* $\operatorname{span}(AC) \cap \operatorname{span}(BC) = \operatorname{span}(C)$.

Proof. The proof follows that of Theorem 3.1, with linear span in place of equality: the first five axioms are linear algebra, and extension and the independence theorem come from the freedom to prescribe the form on a complement. We verify the hypotheses of Fact 2.2. Invariance and symmetry are immediate, and finite character holds because a vector of $\operatorname{span}(AC) \cap \operatorname{span}(BC)$ outside $\operatorname{span}(C)$ is a combination of finitely many elements of A, B and C.

For monotonicity and transitivity, put $Q_1 = \operatorname{span}(AC)$, $Q_2 = \operatorname{span}(BC)$, $Q_3 = \operatorname{span}(BCD)$ and $Q_0 = \operatorname{span}(C)$; these subspaces satisfy $Q_2 \subseteq Q_3$ and $\operatorname{span}(ABC) = Q_1 + Q_2$. It remains to be shown that $Q_1 \cap Q_3 = Q_0$ if and only if $Q_1 \cap Q_2 = Q_0$ and $(Q_1 + Q_2) \cap Q_3 = Q_2$. Suppose that $Q_1 \cap Q_3 = Q_0$. Since $Q_2 \subseteq Q_3$, the intersection $Q_1 \cap Q_2$ is contained in Q_0. If $u_1 + u_2 \in Q_3$ with $u_1 \in Q_1$ and $u_2 \in Q_2$, then the vector u_1 also lies in Q_3, hence in $Q_1 \cap Q_3 = Q_0 \subseteq Q_2$; thus the sum lies in Q_2. Conversely, every $u \in Q_1 \cap Q_3$ lies in $(Q_1 + Q_2) \cap Q_3 = Q_2$, and therefore in $Q_1 \cap Q_2 = Q_0$.

Local character follows from finite dimension. Given a finite tuple a and a set B, the space $\operatorname{span}(a) \cap \operatorname{span}(B)$ is finite, so it is contained in $\operatorname{span}(B_0)$ for some finite $B_0 \subseteq B$. If $u + w \in \operatorname{span}(B)$ with $u \in \operatorname{span}(a)$ and $w \in \operatorname{span}(B_0)$, then the vector u lies in $\operatorname{span}(a) \cap \operatorname{span}(B) \subseteq \operatorname{span}(B_0)$. It follows that $\operatorname{span}(aB_0) \cap \operatorname{span}(B) = \operatorname{span}(B_0)$, which is the relation $a \mathop{\smile\!\!\!\!\!\!|}_{B_0} B$.

For extension, let a and $C \subseteq B$ be given. Replacing a by a maximal subtuple linearly independent over $\operatorname{span}(C)$, we may assume that a is linearly independent over $\operatorname{span}(C)$, since the remaining entries are then determined. Choose a complement B' of $\operatorname{span}(C)$ in $\operatorname{span}(BC)$. By saturation, the alternating space $\operatorname{span}(BC) \oplus \operatorname{span}(x)$ in which x has the values of ω prescribed by $\operatorname{tp}(a/C)$ on $\operatorname{span}(xC)$ and is orthogonal to B' is realized over $\operatorname{span}(BC)$ by a tuple a'; this tuple has the type of a over C and is linearly independent over $\operatorname{span}(BC)$.

The independence theorem is a gluing of functionals. Let M, B, D, a_1 and a_2 be as in Fact 2.2(vii). As before, we may assume that a_1 is linearly independent over M; then so is a_2, and the hypotheses $a_1 \mathop{\smile\!\!\!\!\!\!|}_M B$ and $a_2 \mathop{\smile\!\!\!\!\!\!|}_M D$ make a_1 linearly independent over $\operatorname{span}(MB)$ and a_2 over $\operatorname{span}(MD)$. For each k, the functionals $\omega(a_{1,k}, \cdot)$ on $\operatorname{span}(MB)$ and $\omega(a_{2,k}, \cdot)$ on $\operatorname{span}(MD)$ agree on M, because $\operatorname{tp}(a_1/M) = \operatorname{tp}(a_2/M)$. Since $B \mathop{\smile\!\!\!\!\!\!|}_M D$, their domains meet in $\operatorname{span}(MB) \cap \operatorname{span}(MD) = M$, and the two functionals glue to a linear functional on $\operatorname{span}(MBD)$. Realize over $\operatorname{span}(MBD)$ a tuple a, linearly independent over $\operatorname{span}(MBD)$, with these functionals and with the values of ω on a given by $\operatorname{tp}(a_1/M)$. By quantifier elimination, the type of a over MB is that of a_1, and its type over MD is that of a_2; the relation $a \mathop{\smile\!\!\!\!\!\!|}_M BD$ holds by construction. □

In both structures, then, nonforking independence is the independence of the closure operator, equality for the graph and linear span for the Pauli group, and the independence theorem is the freedom to prescribe the remaining relations, edges or values of the form, on each side separately.

Remark 3.4. The proof of Theorem 3.3 applies over any finite field $\mathbb{F}_q$, in the language with a unary function for multiplication by each scalar and a binary relation for each value of the form. For symmetric or arbitrary bilinear forms, the identification of the limit through symplectic bases and the theorem of Witt is no longer available in characteristic two, but the generic limit exists by free amalgamation, and the verification of Fact 2.2 goes through with the values $\omega(x,x)$ prescribed by the type.

4. Stabilizer codes as a structure: a dictionary

Three notions of model theory, namely Fraïssé limits, forking and the independence theorem, are already at work in quantum error correction under other names, and in this section we identify them. After a brief account of stabilizer codes, in each subsection we state a theorem, prove it, and interpret it twice: once for the physicist, who knows stabilizer codes but not model theory, and once for the logician, who knows model theory but not stabilizer codes. A single example runs through the section: the five-qubit code of Laflamme, Miquel, Paz and Zurek [LMPZ96] and of Bennett, DiVincenzo, Smolin and Wootters [BDSW96], the smallest code that corrects an arbitrary error on one qubit. Table 1 summarizes the dictionary.

Model theory	**Quantum error correction**	
the structure (V, ω)	Pauli strings on countably many qubits, with commutation	Section 3
the age of (V, ω)	the commutation matrices of finite sets of Pauli strings	Proposition 4.2
ultrahomogeneity, automorphisms	Clifford circuits; encoding circuits of stabilizer codes	Theorem 4.3
algebraic closure of a set	the group generated by a set of Pauli strings	Section 3
the type of an error over a set	its syndrome, and whether it lies in the group generated	Theorem 4.5
a forking extension	the error becomes a product of known operators	Theorem 4.5
nonforking over $\emptyset$ of the type over S, for a nonzero label with zero syndrome	an undetectable error that is not harmless	Corollary 4.6
forking over S of the type over $S \cup L$	a logical error	Proposition 4.7
the independence theorem	syndromes of independent checks can be prescribed independently	Proposition 4.8
idealized pure probes	Pauli strings of arbitrary support	Proposition 5.4
probes	Pauli channels	Proposition 5.4
the difference of two coset responses	the projector onto a syndrome sector	Proposition 7.1

Table 1. The Pauli group as a structure: the dictionary of Sections 4, 5 and 7.

4.1. **Stabilizer codes in brief.** We recall what the logician needs (see [Got97]). A *stabilizer group* on n qubits is a group $\mathcal{G}$ of pairwise commuting Pauli strings that does not contain $-I$. Its common eigenspace for the eigenvalue 1 is the *code space*; if $\mathcal{G}$ has $n-k$ independent generators, the code space has dimension 2^k and encodes k qubits. In the labels of Section 3, the group $\mathcal{G}$ corresponds, up to signs, to an isotropic subspace $S \subseteq \mathbb{F}_2^{2n}$, that is, a subspace on which ω vanishes. An error is a Pauli string e; measuring the generators $g_1, \dots, g_{n-k}$ of $\mathcal{G}$ on the corrupted state returns the *syndrome* $(\omega(g_1, e), \dots, \omega(g_{n-k}, e))$, and the decoder guesses

the error from the syndrome. An error with zero syndrome is undetectable. It is harmless if it lies in $\mathcal{G}$, since it then acts trivially on the code space, and it is a *logical error* otherwise. On labels, the errors with zero syndrome form the *centralizer* $S^\perp = \{e \in \mathbb{F}_2^{2n} : \omega(s,e) = 0 \text{ for all } s \in S\}$, which contains S and has dimension $2n-(n-k) = n+k$; a set L of *logical operators* is a set of $2k$ labels with $S^\perp = S \oplus \operatorname{span}(L)$, and the values $\omega(\ell, e)$ for $\ell \in L$ record which logical operator an error of $S^\perp$ applies to the encoded qubits. The *distance* of the code is the least number of qubits on which a logical error acts. A *Clifford unitary* is a unitary operator U that maps Pauli strings to Pauli strings under conjugation, up to a phase; on labels, conjugation by U is a linear map of $\mathbb{F}_2^{2n}$ that preserves ω.

Example 4.1. The five-qubit code has the stabilizer generators

$$g_1 = XZZXI, \quad g_2 = IXZZX, \quad g_3 = XIXZZ, \quad g_4 = ZXIXZ,$$

where $XZZXI$ stands for $X \otimes Z \otimes Z \otimes X \otimes I$, and it encodes one qubit, with logical operators $\bar{X} = XXXXX$ and $\bar{Z} = ZZZZZ$. Each of the fifteen errors that act as X, Y or Z on a single qubit has its own nonzero syndrome, and there are exactly fifteen nonzero syndromes of four bits; for instance, the error XIIII has syndrome $(0,0,0,1)$ and the error $IIZII$ has syndrome $(0,0,1,0)$. The code therefore corrects every error on one qubit, and it has distance 3.

4.2. **Fraïssé limits and Clifford circuits.** In Proposition 4.2 and Theorem 4.3 below, we show that every finite piece of commutation structure occurs among Pauli strings, and that every symmetry between finite pieces is implemented by a Clifford circuit.

Proposition 4.2. *If Θ is a symmetric $d \times d$ matrix over $\mathbb{F}_2$ with zero diagonal, then there are Pauli strings $P_1, \dots, P_d$ on at most d qubits such that P_i and P_j anticommute exactly when $\Theta_{ij} = 1$.*

Proof. The matrix Θ defines an alternating form on $\mathbb{F}_2^d$. Let W_0 be its radical and choose a complement W of W_0; the form is nondegenerate on W, which therefore has a symplectic basis of r pairs, where $2r = \dim W$. Send the r pairs to the labels of X and Z on the first r qubits, and a basis of W_0 to the labels of X on $\dim W_0$ further qubits. This linear map preserves the form, and the images of the standard basis vectors of $\mathbb{F}_2^d$ are the required strings, on $r + \dim W_0 \le d$ qubits. □

Theorem 4.3. *Let W and W' be subspaces of $\mathbb{F}_2^{2n}$ and let $f\colon W \to W'$ be a linear bijection with $\omega(f(v), f(w)) = \omega(v,w)$ for all $v, w \in W$. Then there is a Clifford unitary U on n qubits with $UP_vU^* = \pm P_{f(v)}$ for all $v \in W$. If, moreover, the subspace W is isotropic and the vectors $v_1, \dots, v_m$ form a basis of W, then the unitary U can be chosen with $UP_{v_i}U^* = P_{f(v_i)}$ for all $i \le m$.*

Proof. By the theorem of Witt, the map f extends to a linear bijection τ of $\mathbb{F}_2^{2n}$ that preserves ω. Every such τ is induced by conjugation by a Clifford unitary [CRSS98, DDM03]; let U be one. Since the operator UP_vU^* is a Pauli string up to a phase, Hermitian and with square I, the phase is a sign. For the second statement, let $\varepsilon_i \in \{\pm 1\}$ be the sign with $UP_{v_i}U^* = \varepsilon_i P_{f(v_i)}$. The vectors $f(v_i)$ are linearly independent, so by nondegeneracy there is $b \in \mathbb{F}_2^{2n}$ with $(-1)^{\omega(f(v_i),b)} = \varepsilon_i$ for all i. Conjugation by P_b multiplies $P_{f(v_i)}$ by $(-1)^{\omega(f(v_i),b)}$, so the Clifford unitary P_bU has the required property. □

Corollary 4.4. *Any two stabilizer groups on n qubits with the same number of independent generators are conjugate by a Clifford unitary. In particular, every code that encodes k*

qubits into n has an encoding circuit made of Clifford gates, which maps the trivial code with generators $Z_1, \dots, Z_{n-k}$ onto it.

Proof. Let $g_1, \dots, g_m$ and $g'_1, \dots, g'_m$ be independent generators, with labels v_i and v'_i. Both families span isotropic subspaces, so the linear map $v_i \mapsto v'_i$ preserves ω, and Theorem 4.3 gives a Clifford unitary U with $Ug_iU^* = g'_i$ for all i. The signs of the generators are absorbed as in the proof of Theorem 4.3. □

On countably many qubits, every finite-dimensional subspace of V lies in the labels of the first n qubits for some n. Applying Theorem 4.3 there, and extending by the identity on the remaining qubits, shows that every isometry between finite-dimensional subspaces of (V, ω) extends to an automorphism of (V, ω) implemented by a Clifford circuit on finitely many qubits.

For the physicist or operator theorist. A Fraïssé limit is the unique countable structure that contains every finite configuration of a given kind and in which every symmetry between two finite configurations extends to a symmetry of the whole structure. By Proposition 4.2, every finite pattern of commutation relations occurs among Pauli strings, and by Theorem 4.3, every symmetry between finite pieces extends to a global one, implemented by a Clifford circuit. The Pauli group on countably many qubits is therefore the Fraïssé limit of finite commutation patterns, and Clifford circuits on finitely many qubits implement the symmetries that witness its homogeneity. The existence of encoding circuits for stabilizer codes, usually proved by an explicit construction, is the shadow of this homogeneity: by Corollary 4.4, the five-qubit code is the image of the trivial code with generators Z_1, Z_2, Z_3, Z_4 under a Clifford circuit, because the two generating sets have the same commutation pattern, namely none at all.

For the logician. The structure (V, ω) of Section 3 is the Pauli group modulo phases, and the Clifford group acts on it through the symplectic group, which is the automorphism group of each finite piece $(\mathbb{F}_2^{2n}, \omega)$. Ultrahomogeneity, which in Section 3 came from the theorem of Witt and a back-and-forth argument, is implemented by unitary operators on a Hilbert space, the gates of a quantum computer. Stabilizer codes are finite isotropic subspaces, and Corollary 4.4 is ultrahomogeneity for isotropic subspaces, with the signs of the generators as additional data handled by Lemma 3.2.

4.3. **Forking and logical errors.** In a simple theory, forking is the notion of an extension of a type that adds information. The next theorem computes it for the Pauli group, using the description of nonforking independence in Theorem 3.3.

Theorem 4.5. *Let e be an element of a monster model $\mathbb{M}$ of the theory of (V, ω), and let $D \subseteq B$ be small subsets of $\mathbb{M}$. Then the type $\mathrm{tp}(e/B)$ forks over D if and only if $e \in \mathrm{span}(B) \setminus \mathrm{span}(D)$.*

Proof. By Theorem 3.3, the type $\mathrm{tp}(e/B)$ does not fork over D exactly when $\mathrm{span}(eD) \cap \mathrm{span}(B) = \mathrm{span}(D)$. If $e \in \mathrm{span}(D)$, the spans $\mathrm{span}(eD)$ and $\mathrm{span}(D)$ coincide, and the equality holds. Otherwise the span of eD is $\mathrm{span}(D) \oplus \{0, e\}$, and the equality fails if and only if some vector $e + d$ with $d \in \mathrm{span}(D)$ lies in $\mathrm{span}(B)$, that is, if and only if $e \in \mathrm{span}(B)$. □

By Theorem 4.5, commutation data never fork: the type of e over B records the values $\omega(b, e)$ for $b \in B$, and these values can be prescribed freely without forking. Forking happens only when e becomes a product of the operators in B. In a stabilizer code this criterion

separates the two kinds of undetectable error. The *weight* of an error is the number of qubits on which it acts.

Corollary 4.6. *Let $\mathcal{G}$ be a stabilizer group with set of labels S, and let e be a nonzero label with zero syndrome. Then the type $\mathrm{tp}(e/S)$ forks over $\emptyset$ if and only if e is harmless, that is, $e \in S$. Consequently, the distance of the code is the least weight of a nonzero error with zero syndrome whose type over S does not fork over $\emptyset$.*

Proof. Apply Theorem 4.5 with $D = \emptyset$ and $B = S$; the span of S is S itself, and $\mathrm{span}(\emptyset) = \{0\}$. The second statement is the definition of distance, since a logical error is a nonzero label with zero syndrome outside S. □

In Proposition 4.7 below, we show that the stabilizer group alone does not tell logical errors apart, because they all have the same type over S, and that the logical operators, once known, characterize them by forking.

Proposition 4.7. *Let $\mathcal{G}$ be a stabilizer group with set of labels S, let L be a set of logical operators, and let e be a label with zero syndrome. Then the following hold:*

(a) *the label e is a logical error if and only if $\mathrm{tp}(e/S \cup L)$ forks over S;*
(b) *all logical errors have the same type over S, and this type does not fork over $\emptyset$.*

Proof. (a) Since the syndrome of e is zero, the label e lies in $S^\perp = \mathrm{span}(S \cup L)$. By Theorem 4.5 with $D = S$ and $B = S \cup L$, the type $\mathrm{tp}(e/S \cup L)$ forks over S if and only if $e \in \mathrm{span}(S \cup L) \setminus S = S^\perp \setminus S$, which is the set of logical errors.

(b) Suppose that the label e is a logical error, that is, $e \in S^\perp \setminus S$. By quantifier elimination, the type of e over S is determined by the linear relations of e over S, of which there are none since $e \notin S$, and by the values of ω on $\mathrm{span}(eS)$. These values are fixed, through bilinearity, by the values of ω on S, by $\omega(e, e)$, which is 0 because ω is alternating, and by the values $\omega(s, e)$ for $s \in S$, which are 0 because e lies in $S^\perp$. Hence all logical errors have the same type over S. The label e is nonzero, since the zero label lies in S; as it has zero syndrome and lies outside S, its type over S does not fork over $\emptyset$, by Corollary 4.6. □

In the five-qubit code, the errors $g_1 g_2 = XYIYX$ (up to a phase) and $\bar{X} = XXXXX$ both have zero syndrome. The first lies in the stabilizer group, so its type over S forks over $\emptyset$; it is harmless. The second does not, so its type does not fork over $\emptyset$; it flips the encoded qubit, and it acts on five qubits. The least weight of an error of the second kind is 3, attained for instance by $\bar{X} g_1 = IYYIX$ up to a phase. With $L = \{\bar{X}, \bar{Z}\}$, the logical errors $\bar{X}$, $\bar{Z}$ and $\bar{X} g_1$ have the same type over S; over $S \cup L$ each of them lies in $\mathrm{span}(S \cup L)$, whence its type becomes algebraic and forks over S, as Proposition 4.7(a) predicts.

For the physicist or operator theorist. Forking is the model-theoretic notion of an observation that singles out one of infinitely many alternatives that the data known in advance cannot tell apart and that exclude one another. To be precise, in a simple theory, by a theorem of Kim [Kim98], a condition $\varphi(x, b)$ on an unknown element x forks over a set D of known parameters exactly when, for some $k \geq 2$, the parameter b is the first term of a sequence $(b_j)_{j \in \mathbb{N}}$ that is k-inconsistent for φ (see Definition 2.1) and indiscernible over D. A sequence is indiscernible over D if any two of its increasing finite subsequences of the same length satisfy the same formulas with parameters from D. A type forks over D when one of its conditions does; such a type narrows the possibilities for the element but need not determine it. In the Pauli group a forking type does determine the element. The type of an error over a set B of known operators records the syndrome of the error with respect to B and whether,

and how, the error is a product of operators in B; by Theorem 4.5, this type forks over a subset D of B exactly when the error is a product of operators in B but not of operators in D. Syndromes, by contrast, only narrow: on infinitely many qubits, no finite set of checks determines an error, however many bits it reveals. In a code this criterion separates the two kinds of undetectable error other than the identity (see Corollary 4.6): the type over S of a harmless one, which is a product of stabilizers, forks over the empty set, whereas that of a logical error does not. The distance of the code, its single most important parameter, is therefore the least weight of an undetectable error other than the identity whose type over S does not fork over the empty set. Proposition 4.7 completes the picture: the stabilizer group alone sees all logical errors alike, and each logical error is pinned down as soon as the logical operators join the known operators, since its type over $S \cup L$ forks over S.

For the logician. The theory of (V, ω) has a vector space pregeometry for algebraic closure, and by Theorem 4.5 its forking is the dependence of this pregeometry: the type of a single element over a set forks over a subset exactly when the element is algebraic over the set but not over the subset. Thus the theory has SU-rank one, as does the theory of the Rado graph (see Theorem 3.1), with linear span in place of equality. In finite dimension, the information carried by a syndrome is measured by counting: if $s_1, \dots, s_m$ are linearly independent labels in $\mathbb{F}_2^{2n}$, then for each syndrome $\sigma \in \mathbb{F}_2^m$ exactly 2^{2n-m} labels e satisfy $\omega(s_i, e) = \sigma_i$ for all i, since the map $e \mapsto (\omega(s_i, e))_{i \le m}$ is linear and onto. This count records codimension, which forking ignores. The logical errors of a code are the elements of the centralizer of S outside S, and by Proposition 4.7 they share one type over S. Its nonforking extensions to $S \cup L$ are realized in (V, ω) by labels outside $\operatorname{span}(S \cup L)$ that commute with S, for instance by errors on qubits that the code does not touch; the logical errors realize its forking extensions, which become algebraic once the logical operators are added.

4.4. **The independence theorem and the gluing of syndromes.** The independence theorem of Kim and Pillay says that two types that do not fork can be amalgamated. In the Pauli group it becomes a statement about syndromes.

Proposition 4.8. *Let C, B and D be subsets of $\mathbb{F}_2^{2n}$ with $\operatorname{span}(BC) \cap \operatorname{span}(DC) = \operatorname{span}(C)$, and let ξ_1 and ξ_2 be linear functionals on $\operatorname{span}(BC)$ and on $\operatorname{span}(DC)$ that agree on $\operatorname{span}(C)$. Then there is a label $e \in \mathbb{F}_2^{2n}$ with $\omega(s, e) = \xi_1(s)$ for $s \in \operatorname{span}(BC)$ and $\omega(s, e) = \xi_2(s)$ for $s \in \operatorname{span}(DC)$.*

Proof. Since ξ_1 and ξ_2 agree on the intersection of their domains, there is a linear functional ξ on $\operatorname{span}(BC) + \operatorname{span}(DC)$ extending both. Extend ξ to $\mathbb{F}_2^{2n}$. Because ω is nondegenerate, every linear functional on $\mathbb{F}_2^{2n}$ has the form $\omega(\cdot, e)$ for a unique e. □

In the five-qubit code, take $C = \emptyset$, $B = \{g_1, g_2\}$ and $D = \{g_3, g_4\}$. The spans of B and D meet only in 0, so any two bits of syndrome on g_1, g_2 and any two on g_3, g_4 are realized together by one error; by Example 4.1, even by an error on a single qubit, for instance $YIIII$ for the syndrome $(1, 0 \,|\, 1, 1)$. If D is replaced by $\{g_1 g_2\}$, the hypothesis fails, and so does the conclusion: the syndrome of $g_1 g_2$ is the sum of those of g_1 and g_2.

For the physicist or operator theorist. The independence theorem is the rule that lets partial information be combined: two sets of checks that are independent given their common part impose no constraint on each other, so their syndromes can be prescribed separately and are realized by a single error. The point of the theorem of Kim and Pillay is that this gluing rule, together with the other properties of Fact 2.2, characterizes the simple theories [KP97].

In Section 6 we show that the rule has no counterpart once preparations are mixed: for two halves of a block that meet in a quarter of it, the conditions "the probe answers 1 on this half and 0 on the rest of the block" and "on that half" are each consistent but cannot hold together, and infinitely many such conditions produce the tree property of the second kind.

For the logician. Proposition 4.8 is the independence theorem of Theorem 3.3 in finite dimension, over an arbitrary closed set rather than over a model: the gluing of functionals that agree on $\operatorname{span}(BC) \cap \operatorname{span}(DC) = \operatorname{span}(C)$ is the whole argument. Its physical content is that the syndromes of a set of checks are determined by a linear functional on the span of the checks, and the consistency of prescribed syndromes is therefore a matter of linear algebra on that span; the surjectivity of the syndrome map of linearly independent checks is Lemma 3.2.

5. Observables, mixtures and dividing

We now transport Definition 2.1 to the responses of an observable, in the form used in the second paper, and introduce the observables of the two paradigms. Their probes, we find, are of different kinds (see Proposition 5.4 and Lemma 5.5): the probes of the Pauli observable are the Pauli channels, and its idealized pure probes are the Pauli strings of infinite support; the pure probes of the Rado observable are all sets of neighbors.

5.1. **Diagonal observables.** Let A and B be countable sets and let $\varphi\colon A \times B \to \{0,1\}$ be a function. The *diagonal observable* of φ is the operator

$$C_\varphi = \sum_{a \in A,\, b \in B} \varphi(a,b)\, |a\rangle\langle a| \otimes |b\rangle\langle b| \quad \text{on } \ell^2(A) \otimes \ell^2(B).$$

In plain words, Alice's register holds a label a, Bob's a label b, and the observable reads $\varphi(a,b)$. A density operator ρ of Alice, that is, a positive operator of trace one on $\ell^2(A)$ that describes her preparation, enters only through its diagonal, which is a probability measure μ on A: the preparation draws a at random with law μ. The slice of C_φ by ρ, the operator on Bob's space obtained by averaging C_φ over Alice's preparation, is the diagonal operator of multiplication by

$$b \longmapsto \sum_{a \in A} \mu(a)\varphi(a,b).$$

The probe algebra of the second paper is the unital C*-algebra generated by these slices; it is the commutative C*-algebra of bounded functions on B generated by the constants and the indicators $b \mapsto \varphi(a,b)$, $a \in A$.

Lemma 5.1. *Let $K_\varphi \subseteq \{0,1\}^A$ be the closure of the set of functions $\varphi(\cdot,b)$, $b \in B$. The probe algebra of C_φ is isomorphic to $C(K_\varphi)$, the probe space is the space $\mathcal{P}(K_\varphi)$ of Radon probability measures on K_φ, and the response of the mixture μ is the affine continuous function*

$$r_\mu(\nu) = \int_{K_\varphi} \sum_{a \in A} \mu(a)\lambda(a)\, d\nu(\lambda) \qquad (\nu \in \mathcal{P}(K_\varphi)).$$

Proof. The unital C*-algebra generated by commuting projections is the algebra of continuous functions on the Stone space of the Boolean algebra they generate. The Boolean algebra generated by the sets $\{b : \varphi(a,b) = 1\}$ has as Stone space the closure of the image of B under $b \mapsto \varphi(\cdot,b)$, which is K_φ. States of $C(K_\varphi)$ are Radon probability measures by the Riesz representation theorem, and the formula for r_μ is the evaluation of the state ν at the slice. □

The pure probes are the point masses δ_λ, $\lambda \in K_\varphi$; a pure probe is an *idealized label* of Bob, recording which of Alice's labels it answers with 1. Actual labels $b \in B$ are dense among them, since K_φ is the closure of the set of functions $\varphi(\cdot, b)$. For a label b, we write $r_\mu(b)$ for the response at the pure probe of $\varphi(\cdot, b)$.

5.2. **The structure of an observable.** To speak of the theory of an observable, we regard it as a structure of real-valued logic. For a diagonal observable C_φ, let $\mathcal{S}_\varphi$ be the two-sorted structure whose sorts are Alice's mixtures, the probability measures on A, and Bob's probes, the space $\mathcal{P}(K_\varphi)$, and whose only predicate is the response $R(\mu, \nu) = r_\mu(\nu)$, with values in $[0, 1]$. As in the second paper, we work in the real-valued logic of Keisler [Kei23], in which no metric is given a priori; in an ultrapower, two elements are identified when every atomic formula takes the same value at them. We say that C_φ is *simple* if the real-valued theory of $\mathcal{S}_\varphi$ is simple in the sense of Ben Yaacov [BY03]. In an elementary extension of $\mathcal{S}_\varphi$, the new elements of the sort of preparations are idealized preparations of Alice, and the tree properties of a formula are witnessed there. When a tree property is witnessed in $\mathcal{S}_\varphi$ itself, in the sense of Definition 5.2 below, we say that actual preparations witness it.

5.3. **Dividing for families of functions.** The responses of the second paper form a family of real functions on the probe space, and the independence property was defined for such a family. We do the same for the tree properties. The conditions are inequalities with a gap, which is what makes them survive in elementary extensions of the structure $\mathcal{S}_\varphi$.

Definition 5.2. Let Y be a set and $\mathcal{R}$ a family of real functions on Y.

(1) A sequence $(r_j)_{j\in\mathbb{N}}$ in $\mathcal{R}$ is *uniformly k-inconsistent* if there are reals $\alpha < \beta$ such that each r_j takes a value $\geq \beta$ somewhere on Y, and for all $j_1 < \cdots < j_k$ no $y \in Y$ satisfies $r_{j_\ell}(y) > \alpha$ for $\ell = 1, \ldots, k$.

(2) The family $\mathcal{R}$ has the *tree property of the second kind* (TP_2) if there are $k \geq 2$, reals $\alpha < \beta$ and an array $(r_{ij})_{i,j\in\mathbb{N}}$ in $\mathcal{R}$ such that every row is uniformly k-inconsistent with the constants α and β, and for every $f\colon \mathbb{N} \to \mathbb{N}$ and every n there is $y \in Y$ with $r_{if(i)}(y) \geq \beta$ for all $i < n$.

The tree property of Definition 2.1, its variant of the first kind, and TP_2 all require infinite sequences of instances that are k-inconsistent: the children of a node, or a row. When these notions are transported to a family of functions as in Definition 5.2, those sequences become uniformly k-inconsistent. A family without uniformly k-inconsistent sequences therefore does not have the tree property of the second kind. For the response of an observable, Definition 5.2 is the tree property of the second kind of the atomic formula, in the real-valued logic of the second paper; the gap between α and β is what the definitions of dividing in real-valued logic require [BY03].

5.4. **The Pauli observable.** The *Pauli observable* is the diagonal observable of ω on $V \times V$: Alice and Bob each hold a Pauli label, and the observable reports whether the two Pauli strings anticommute. We call Alice's label the check and Bob's label the error; the observable then reports one syndrome bit. For a mixture μ, the response at an error e is the probability that a check drawn with law μ anticommutes with e. By Lemma 3.2, the observable has the independence property already on pure preparations.

Lemma 5.3. *For the Pauli observable, the space K_ω is the group V^* of all linear functionals $V \to \mathbb{F}_2$.*

Proof. Each $\omega(\cdot, e)$ is linear, and the linear functionals form a closed subset of $\{0, 1\}^V$. Conversely, a basic neighborhood of a linear functional λ prescribes the values of λ on finitely

many vectors, which may be taken linearly independent, and Lemma 3.2 gives e with $\omega(\cdot, e)$ in the neighborhood. □

The functionals of V^* have a direct physical meaning. Let $\hat{V} = \prod_{k\in\mathbb{N}} \mathbb{F}_2^2$ be the group of labels of Pauli strings of arbitrary support, with the product topology; it contains V as the strings of finite support. For $a \in V$ and $u \in \hat{V}$, the value $\omega(a, u)$ is defined by the formula of Section 3, which is a finite sum because a has finite support. For p with $0 < p \leq 1$, let ν_p be the *depolarizing noise* of rate p: the product measure on $\hat{V}$ under which the labels on distinct qubits are independent, each equal to $(0, 0)$ with probability $1 - p$ and to each of $(1, 0)$, $(0, 1)$ and $(1, 1)$ with probability $p/3$. A *Pauli channel* on countably many qubits is a Radon probability measure on $\hat{V}$, the law of a random error applied to the qubits.

Proposition 5.4. *Let $\hat{V}$ be the group of labels of Pauli strings of arbitrary support, and for $0 < p \leq 1$ let ν_p be the depolarizing noise of rate p. Then the following hold:*

(a) *the map $u \mapsto \omega(\cdot, u)$ is an isomorphism of topological groups from $\hat{V}$ onto V^*, which carries V onto the functionals of the actual errors;*
(b) *the probes of the Pauli observable are the Pauli channels on countably many qubits, and the response of a mixture μ at a Pauli channel ν is the probability that a check drawn with law μ anticommutes with an independent error drawn with law ν;*
(c) *the depolarizing noise ν_p is concentrated on errors of infinite support, and for a label $a \in V$ that acts nontrivially on w qubits we have*

$$r_{\delta_a}(\nu_p) = \tfrac{1}{2}\big(1 - (1 - \tfrac{4p}{3})^w\big).$$

Proof. (a) The map is a homomorphism. Write X_k and Z_k for the labels of the strings X and Z on the qubit k. Then the values $\omega(X_k, u)$ and $\omega(Z_k, u)$ are the two coordinates of u at the qubit k, so the map is injective, and every $\lambda \in V^*$ is the image of the vector $u = (x, z)$ with $z_k = \lambda(X_k)$ and $x_k = \lambda(Z_k)$ for every k, because λ and $\omega(\cdot, u)$ are linear and agree on the basis $\{X_k, Z_k : k \in \mathbb{N}\}$ of V. The value $\omega(a, u)$ depends on finitely many coordinates of u, so the map is continuous, and a continuous bijection between compact Hausdorff spaces is a homeomorphism.

(b) By Lemma 5.1, Lemma 5.3 and (a), the probes are the Radon probability measures on $\hat{V}$, and $r_\mu(\nu) = \int \sum_a \mu(a)\,\omega(a, u)\,d\nu(u)$, which is the probability stated, by Fubini's theorem.

(c) Under ν_p, the events "the error acts nontrivially on the qubit k" are independent and have probability p, so, by the lemma of Borel and Cantelli, almost every error acts nontrivially on infinitely many qubits. At each of the w qubits where P_a acts nontrivially, it anticommutes with two of the three nontrivial Pauli matrices, hence with the error with probability $2p/3$, independently of the other qubits. The strings anticommute exactly when an odd number of these w events occur, and this has probability $\frac{1}{2}\big(1 - (1 - \frac{4p}{3})^w\big)$. □

By Proposition 5.4, the noise of a memory on countably many qubits is a probe of the Pauli observable whose support consists of idealized errors: with probability one, the error acts on infinitely many qubits and is not an actual label. Alice's checks see such an error through finitely many qubits at a time, and the response of a check of weight w tends to $\frac{1}{2}$, the value of a fair coin, as w grows. In what follows we identify V^* with $\hat{V}$ and call its elements *idealized errors*. For $a \in V$, let χ_a be the character of V^* given by $\chi_a(\lambda) = (-1)^{\lambda(a)}$. Alice's pure responses are the characters of this compact group up to an affine change: we have $\lambda(a) = (1 - \chi_a(\lambda))/2$.

5.5. **The Rado observable.** The *Rado observable* is the diagonal observable of E on $\Gamma \times \Gamma$: Alice and Bob each hold a site of a random network of couplings, and the observable reports whether the two sites are coupled. By the extension property, it has the independence property already on pure preparations. We say that a diagonal observable has *freely independent* pure responses when $K_\varphi = \{0,1\}^A$, that is, when every pattern of answers to all of Alice's labels is an idealized label of Bob.

Lemma 5.5. *For the Rado observable, the space K_E is the whole Cantor cube $\{0,1\}^\Gamma$. Hence the Rado observable has freely independent pure responses, while the Pauli observable does not.*

Proof. A basic open subset of $\{0,1\}^\Gamma$ prescribes adjacency to the vertices of a finite set Γ_1 and nonadjacency to those of a finite set Γ_0, and the extension property gives a vertex outside $\Gamma_1 \cup \Gamma_0$ with that pattern. By Lemma 5.3, the space K_ω is the group V^*, which is a proper subset of $\{0,1\}^V$. □

The pure probes of the Rado observable are arbitrary sets of neighbors, that is, arbitrary subsets of Γ, and its probes are random sets of neighbors. The response of a mixture μ at a probe ν is the probability that a site drawn with law μ lies in an independent set of neighbors drawn with law ν; it depends on ν only through the marginals $\nu\{\lambda : \lambda(a) = 1\}$, the probabilities that the individual sites a are neighbors. The Pauli observable sees more of a probe: its pure responses are, up to an affine change, the characters of the compact group of idealized errors, and products of characters are again characters. In Sections 7 and 8 we show what this difference does to the tree property.

6. The theory of mixed preparations has the tree property of the second kind

This section is devoted to Theorem 1.1(C), and its idea is a comparison of two averages. Let J be a finite set of labels and H a half of J. The difference between the response to the uniform mixture on H and the response to the uniform mixture on $J \setminus H$ reaches its maximum 1 exactly when the probe answers 1 on H and 0 on $J \setminus H$. Two halves that meet in a quarter of J cannot both be nearly maximized, while the independence property supplies a pure probe with any prescribed answers on finitely many disjoint blocks. These three facts, which we prove in the proof of Theorem 6.1, together give the tree.

Theorem 6.1 (Mixed preparations and the tree property of the second kind). *Let C_φ be a diagonal observable, and assume that there are distinct labels $a_0, a_1, \dots$ in A such that for all finite disjoint $N_1, N_0 \subseteq \mathbb{N}$ there is a label $b \in B$ with $\varphi(a_n, b) = 1$ for $n \in N_1$ and $\varphi(a_n, b) = 0$ for $n \in N_0$. Then the affine formula*

$$\psi(x; y_1, y_2) = R(y_1, x) - R(y_2, x),$$

with x of the sort of probes and y_1, y_2 of the sort of preparations, has the tree property of the second kind in the real-valued theory of $\mathcal{S}_\varphi$, with $k = 2$, $\alpha = 1/2$ and $\beta = 1$. In particular, a simple diagonal observable has no independence property.

Proof. We first build finite arrays in $\mathcal{S}_\varphi$ itself and then pass to an elementary extension by compactness.

Let $N \geq 1$ be given, and choose t with $2^t > N$. Partition $2^t N$ of the labels a_n into N blocks $J_0, \dots, J_{N-1}$ of size 2^t, and identify each block with $\mathbb{F}_2^t$. Choose distinct nonzero vectors $w_0, \dots, w_{N-1}$ in $\mathbb{F}_2^t$, and for $i, j < N$ let $H_{ij} \subseteq J_i$ be the set of $v \in \mathbb{F}_2^t$ with $w_j \cdot v = 1$.

Each H_{ij} has 2^{t-1} elements, and for $j \neq j'$ the set $H_{ij} \cap H_{ij'}$ has 2^{t-2} elements, since w_j and $w_{j'}$ are linearly independent. Let μ^+_{ij} and μ^-_{ij} be the uniform probability measures on H_{ij} and on $J_i \setminus H_{ij}$.

For a probe ν, write $\pi_a(\nu) = \nu\{\lambda : \lambda(a) = 1\} \in [0,1]$ for its marginal at the label a; by Lemma 5.1, the response is then $R(\mu,\nu) = \sum_a \mu(a)\pi_a(\nu)$. With $m = 2^t$, we have

$$\psi(\nu; \mu^+_{ij}, \mu^-_{ij}) = \frac{2}{m}\Big(\sum_{a \in H_{ij}} \pi_a(\nu) - \sum_{a \in J_i \setminus H_{ij}} \pi_a(\nu)\Big) \leq 1.$$

For $j \neq j'$ in the same row, adding the two values gives

$$\frac{4}{m}\Big(\sum_{a \in H_{ij} \cap H_{ij'}} \pi_a(\nu) - \sum_{a \in J_i \setminus (H_{ij} \cup H_{ij'})} \pi_a(\nu)\Big) \leq \frac{4}{m} \cdot \frac{m}{4} = 1,$$

so the two values are never both greater than $1/2$. For every $f\colon N \to N$, the hypothesis applied to the finite disjoint sets $\bigcup_i H_{if(i)}$ and $\bigcup_i (J_i \setminus H_{if(i)})$ gives a label b, and at the pure probe of $\varphi(\cdot, b)$ every value $\psi(\cdot; \mu^+_{if(i)}, \mu^-_{if(i)})$ with $i < N$ equals 1.

Thus, for every N, the structure $\mathcal{S}_\varphi$ satisfies the following conditions on $2N^2$ preparations $y^\pm_{ij}$: for every $f\colon N \to N$, the value of $\sup_x \min_{i<N} \psi(x; y^+_{if(i)}, y^-_{if(i)})$ is 1; and for all i and $j \neq j'$, the value of $\sup_x \min\big(\psi(x; y^+_{ij}, y^-_{ij}), \psi(x; y^+_{ij'}, y^-_{ij'})\big)$ is at most $1/2$. These are finitely many closed conditions for each N, and the conditions for N are inherited by subarrays. By compactness, an $\aleph_1$-saturated elementary extension of $\mathcal{S}_\varphi$ contains an array $(y^\pm_{ij})_{i,j \in \mathbb{N}}$ satisfying them for all N; by saturation, the suprema along paths are attained. This is the tree property of the second kind, which is incompatible with simplicity in real-valued logic [BY03].

It remains to be shown that a simple diagonal observable has no independence property. The hypothesis of the theorem is the independence property of the pure responses, in the form used in the second paper: actual labels are dense in K_φ, and a finite pattern of answers to the labels a_n prescribes a clopen subset of K_φ, so the hypothesis says exactly that the functions $\lambda \mapsto \lambda(a_n)$ form an independent sequence on K_φ. For a diagonal observable, the pure and the mixed responses have the independence property together, since a mixed response is an average of pure ones. Indeed, if the pure responses have no independent sequence, then, by the dichotomy of Rosenthal, every sequence of them has a subsequence that converges pointwise on K_φ, and this subsequence is weakly Cauchy in $C(K_\varphi)$ by dominated convergence; by the theorem of Stegall, as in the second paper, every sequence in the closed absolutely convex hull of the pure responses, which contains the mixed responses, then has a weakly Cauchy subsequence as well. Consequently, if a diagonal observable has the independence property, the hypothesis holds, and the real-valued theory of $\mathcal{S}_\varphi$ has the tree property of the second kind. □

The tree of Theorem 6.1 lives in an elementary extension. In the next two sections we show that for the Pauli observable it can already be found among actual preparations, and that for the Rado observable it cannot.

7. The Pauli observable: syndrome sectors and the first check to fire

We turn to Theorem 1.1(D): for the Pauli observable, actual preparations witness the tree property of the second kind. The tool is the formula for the projector onto a syndrome sector as an average over a group of checks, which we first state for responses. The idea of the theorem is then to let a single mixture ask a question about many checks at once. Let a

block of independent checks $g_0, g_1, \dots$ be given, and let Alice draw a Pauli string uniformly from the coset $g_j + \operatorname{span}(g_0, \dots, g_{j-1})$. In the proof of Theorem 7.2 we show that if the checks before g_j are silent on Bob's error and the check g_j fires, every string in the coset anticommutes with the error, and that if one of the earlier checks fires, exactly half of them do. Two different values of j cannot both describe the first check to fire, and the syndromes of disjoint blocks of linearly independent checks can be prescribed separately (see Lemma 3.2).

7.1. **Syndrome sectors.** Let $g_1, \dots, g_d \in V$ be linearly independent checks, with span G, and let a syndrome $\sigma \in \mathbb{F}_2^d$ be given. We regard σ as the linear functional on G with $\sigma(g_i) = \sigma_i$, and we call the set $\Sigma_\sigma = \{\lambda \in V^* : \lambda(g_i) = \sigma_i \text{ for } i \le d\}$ of idealized errors with syndrome σ its *syndrome sector*. For $s \in G$ we write r_s for the pure response r_{δ_s}.

Proposition 7.1. *The indicator of the syndrome sector Σ_σ satisfies, at every pure probe $\lambda \in V^*$,*

$$\mathbf{1}_{\Sigma_\sigma}(\lambda) = 2^{-d} \sum_{s \in G} (-1)^{\sigma(s)} \chi_s(\lambda) = 2^{-d} \sum_{s \in G} (-1)^{\sigma(s)} - 2^{1-d} \sum_{s \in G} (-1)^{\sigma(s)} r_s(\lambda).$$

Consequently, if $\sigma \neq 0$, then $\mathbf{1}_{\Sigma_\sigma} = r_{\mu_\sigma^1} - r_{\mu_\sigma^0}$ on pure probes, where μ_σ^1 and μ_σ^0 are the uniform probability measures on $\{s \in G : \sigma(s) = 1\}$ and on $\{s \in G : \sigma(s) = 0\}$; and $\mathbf{1}_{\Sigma_0} = 1 - 2r_{\mu_G}$, where μ_G is the uniform probability measure on G. In particular, for every probe ν, the probability that an error drawn from the Pauli channel ν has syndrome σ is $r_{\mu_\sigma^1}(\nu) - r_{\mu_\sigma^0}(\nu)$ if $\sigma \neq 0$, and $1 - 2r_{\mu_G}(\nu)$ if $\sigma = 0$.

Proof. The map $s \mapsto (-1)^{\sigma(s)} \chi_s(\lambda) = (-1)^{(\sigma+\lambda)(s)}$ is a character of the finite group G; its sum over G is 2^d if the linear functional $\sigma + \lambda$ vanishes on G, that is, if $\lambda \in \Sigma_\sigma$, and 0 otherwise, since a nonzero functional on G takes the value 1 on exactly half of G. This is the first equality, and the second follows from $\chi_s = 1 - 2r_s$. If $\sigma \neq 0$, the first sum on the right vanishes, and the two level sets of σ have 2^{d-1} elements each; grouping the second sum by the value of σ gives $r_{\mu_\sigma^1} - r_{\mu_\sigma^0}$. If $\sigma = 0$, the first sum is 2^d, and the second is $2^d r_{\mu_G}$. Both sides of each identity are affine in the probe, and integration against ν, together with Proposition 5.4(b), yields the last statement. □

When the checks commute pairwise and carry signs, the operator $2^{-d} \sum_{s \in G} (-1)^{\sigma(s)} P_s$ is the projector onto the syndrome sector σ of the code [Got97], and Proposition 7.1 gives the probability of the outcome σ when the checks are measured on a code state corrupted by the channel ν. The proposition requires no commutation: the checks need only be linearly independent. The coefficients of the indicator, as an affine combination of Alice's responses, have total absolute value at most $1 + 2 = 3$, whatever the number d of checks.

7.2. **The first check to fire.** The next theorem arranges the syndrome sectors of nested blocks of checks in an array that witnesses the tree property of the second kind.

Theorem 7.2 (The first check to fire). *Let $(g_{im})_{i,m\in\mathbb{N}}$ be linearly independent vectors of V, put $W_{ij} = \operatorname{span}(g_{im} : m < j)$, and let μ_{ij} be the uniform probability measure on the coset $g_{ij} + W_{ij}$. For the Pauli observable, the following hold:*

(a) *for every $f\colon \mathbb{N} \to \mathbb{N}$ and every n, there is an error $e \in V$ with $r_{\mu_{if(i)}}(e) = 1$ for all $i < n$, and there is an idealized error $\lambda \in V^*$ with $r_{\mu_{if(i)}}(\lambda) = 1$ for all i;*

(b) *for every i, every $j \neq j'$ and every probe $\nu \in \mathcal{P}(V^*)$, we have $\min\big(r_{\mu_{ij}}(\nu), r_{\mu_{ij'}}(\nu)\big) \leq 2/3$, and the bound $2/3$ is attained.*

Consequently, the responses of Alice have the tree property of the second kind, with $k = 2$, $\alpha = 2/3$ and $\beta = 1$.

Proof. We first compute the response of μ_{ij} at a pure probe $\lambda \in V^*$. If λ does not vanish on W_{ij}, then the functional λ takes the value 1 on exactly half of the coset $g_{ij} + W_{ij}$, since a nonzero functional on a finite $\mathbb{F}_2$-space is balanced, and translation by g_{ij} either preserves or exchanges its level sets. If λ vanishes on W_{ij}, then the functional λ is constant on the coset, with value $\lambda(g_{ij})$. Hence

$$r_{\mu_{ij}}(\lambda) = \begin{cases} 1 & \text{if } \lambda(g_{im}) = 0 \text{ for } m < j \text{ and } \lambda(g_{ij}) = 1, \\ 1/2 & \text{if } \lambda(g_{im}) = 1 \text{ for some } m < j, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

In words, the response is 1 exactly when g_{ij} is the first check of block i to fire.

(a) Let f and n be given. The conditions $\lambda(g_{im}) = 0$ for $m < f(i)$ and $\lambda(g_{if(i)}) = 1$, for $i < n$, concern finitely many linearly independent vectors, so Lemma 3.2 gives an error e with $\omega(\cdot, e)$ satisfying them, and by (1) the response $r_{\mu_{if(i)}}(e)$ equals 1 for all $i < n$. The conditions for all i together define a nonempty closed subset of V^*, since the vectors g_{im} are linearly independent; any λ in it works.

(b) By symmetry, we may assume that $j < j'$. By (1), the pair $\big(r_{\mu_{ij}}(\lambda), r_{\mu_{ij'}}(\lambda)\big)$ at a pure probe takes one of the values

$$(1, \tfrac{1}{2}), \quad (\tfrac{1}{2}, \tfrac{1}{2}), \quad (0, \tfrac{1}{2}), \quad (0, 1), \quad (0, 0).$$

Indeed, if the first check of block i to fire among $g_{i0}, \dots, g_{ij'}$ has index m, the pair is $(\frac{1}{2}, \frac{1}{2})$ for $m < j$, $(1, \frac{1}{2})$ for $m = j$, $(0, \frac{1}{2})$ for $j < m < j'$ and $(0, 1)$ for $m = j'$; if none fires, the pair is $(0, 0)$. Responses are affine in the probe, so at a mixed probe ν the pair lies in the convex hull of these five points. On that hull the minimum of the two coordinates is at most $2/3$, with equality only at $\frac{2}{3}(1, \frac{1}{2}) + \frac{1}{3}(0, 1)$, which is the pair of the probe $\frac{2}{3}\delta_\lambda + \frac{1}{3}\delta_{\lambda'}$ with g_{ij} firing first under λ and $g_{ij'}$ under λ'.

Statements (a) and (b) are the two clauses of Definition 5.2 for the array $(r_{\mu_{ij}})$, with $k = 2$, $\alpha = 2/3$ and $\beta = 1$. □

Remark 7.3. For qudits of prime dimension p, the labels form a symplectic space over the field $\mathbb{F}_p$ with p elements, and the response of a label at an error is 1 when their commutation phase is nontrivial and 0 otherwise. The proof of Theorem 7.2 adapts as follows. A nonzero functional on a finite $\mathbb{F}_p$-space takes each value equally often, so in (1) the response $\frac{1}{2}$ becomes $\vartheta = \frac{p-1}{p}$. For $j < j'$, the pair of responses at a pure probe takes one of the values $(1, \vartheta)$, (ϑ, ϑ), $(0, \vartheta)$, $(0, 1)$ and $(0, 0)$. The point (ϑ, ϑ) lies below the segment from $(1, \vartheta)$ to $(0, 1)$, since $\vartheta < 1 - \vartheta + \vartheta^2$, and on that segment the two coordinates are equal at the point with first coordinate $\frac{p}{p+1}$. Hence the bound $\frac{2}{3}$ of Theorem 7.2(b) becomes $\frac{p}{p+1}$, and it is attained.

In the language of Proposition 7.1, the event "the check g_{ij} fires first in block i" is the syndrome sector of the checks $g_{i0}, \dots, g_{ij}$ with syndrome $(0, \dots, 0, 1)$, whose two level sets are the coset $g_{ij} + W_{ij}$ and the group W_{ij}. Hence its indicator is $r_{\mu_{ij}} - r_{\mu^0_{ij}}$ on pure probes, where μ^0_{ij} is the uniform probability measure on W_{ij}. Theorem 7.2 uses only linear independence; when the checks commute pairwise, each W_{ij} is the set of labels of a stabilizer group, and μ_{ij} draws uniformly from a coset of it. The simplest instance is the repetition code.

Example 7.4. Index the qubits by pairs $(i, m) \in \mathbb{N}^2$, so that each block i is a chain of qubits, and let $g_{im} = Z_{i,m} Z_{i,m+1}$ be the check of the bit-flip repetition code on the neighboring qubits m and $m+1$ of the chain i. These checks commute pairwise and are linearly independent. For a bit flip $e = \prod X_{i,m}^{x_{i,m}}$, we have $\omega(g_{im}, e) = x_{i,m} + x_{i,m+1}$, which is 1 if and only if e has a domain wall between the qubits m and $m+1$ of the chain i. By (1), the response of μ_{ij} at e is 1 exactly when the first domain wall of e on the chain i lies between the qubits j and $j+1$. Theorem 7.2 then reads as follows: the chains are independent code blocks of a memory, an error cannot have two first domain walls on the same chain, and the positions of the first domain walls on finitely many chains can be prescribed independently by a single error. The bit flips on the chain i with zero syndrome are the identity and the flip X of the whole chain, which acts on infinitely many qubits: the logical bit flip of an infinite chain is an idealized error, an element of $\hat{V} \setminus V$.

Remark 7.5. The mechanism behind Theorem 7.2 is Fourier analysis on the compact group V^*: since the characters of V^* separate points and are closed under multiplication, the linear span of Alice's responses and the constants is dense in $C(V^*)$ by the Stone–Weierstrass theorem; it is an algebra, because $2r_a r_b = r_a + r_b - r_{a+b}$ for the pure responses r_a. By Proposition 7.1, the indicators of syndrome sectors, which are functions of supremum norm one, lie in this span with coefficients of total absolute value at most 3. On the pure labels, by contrast, a formula of (V, ω) speaks about a bounded number of checks, and Theorem 3.3 shows that no formula has the tree property.

Theorem 6.1 shows that the real-valued theory of $\mathcal{S}_\omega$ is not simple, although the classical theory of (V, ω) is; Theorem 7.2 places the tree in $\mathcal{S}_\omega$ itself, with actual preparations as parameters and the atomic formula as the formula.

8. The Rado observable: marginals and idealized preparations

Theorem 1.1(E), which we prove in this section, concerns the Rado observable. By Theorem 6.1, the Rado observable has the tree property of the second kind, as the Pauli observable does. In Theorem 8.1 and Corollary 8.2 we show that its tree lives among idealized preparations, because conditions on marginals given by actual mixtures are compatible on disjoint blocks. After Corollary 8.2 we note that the same holds for the random bipartite graph, the Fraïssé limit of the finite bipartite graphs with the two sides as two sorts, which has exactly the shape of a bipartite observable: Alice's labels form one side, and Bob's the other.

8.1. **Affine combinations of responses.** For an observable with freely independent pure responses, the response of a mixture depends on a probe only through its marginals $\pi_a(\nu) = \nu\{\lambda : \lambda(a) = 1\}$, as for the Rado observable in Section 5.5, and every vector of marginals in $[0,1]^A$ is attained by a product measure. A bounded affine combination of responses is therefore a function

$$h_{c,\gamma}(\nu) = \gamma + \sum_{a \in A} c(a)\, \pi_a(\nu), \qquad c \in \ell^1(A),$$

and its supremum over the probe space is $\gamma+\sum_a c(a)^+$, attained at the pure probe $\lambda = \mathbf{1}_{\{c>0\}}$. Two such functions can conflict only through the labels on which both put weight. In the proof of Theorem 8.1 below we show that an infinite sequence of them must, after a gliding hump, put its weight on disjoint blocks of labels, apart from a common part; for every k, a single pure probe then nearly maximizes k of them.

Theorem 8.1 (Compatibility of affine conditions). *Let C_φ be a diagonal observable with freely independent pure responses, and let $\kappa > 0$ be given. Let $\mathcal{R}_\kappa$ be the family of functions $\gamma + \sum_{i<n} c_i r_{\mu_i}$ on $\mathcal{P}(\{0,1\}^A)$, where $n \in \mathbb{N}$, the μ_i are mixtures of Alice and $|\gamma| + \sum_i |c_i| \le \kappa$. Then the family $\mathcal{R}_\kappa$ contains no uniformly k-inconsistent sequence, for any k. In particular, no tree property is witnessed in $\mathcal{R}_\kappa$.*

Proof. Let (h_j) be a sequence in $\mathcal{R}_\kappa$ and let $\alpha < \beta$ be such that each h_j takes a value $\ge \beta$. We show that for every k there are $j_1 < \cdots < j_k$ and a pure probe λ with $h_{j_\ell}(\delta_\lambda) > \alpha$ for all ℓ.

Write $h_j = h_{c_j,\gamma_j}$ with $c_j = \sum_i c_{j,i}\mu_{j,i}$, whence $\|c_j\|_1 \le \kappa$. Since A is countable, a diagonal argument gives a subsequence, still denoted (c_j), that converges pointwise on A to some c; by Fatou's lemma, the limit satisfies $\|c\|_1 \le \kappa$. Put $\eta = (\beta - \alpha)/5$ and choose a finite set $F \subseteq A$ with $\|c \restriction (A \setminus F)\|_1 < \eta$. Put $F_0 = F$, and choose inductively indices $j_1 < \cdots < j_k$ and finite sets $F_0 \subseteq F_1 \subseteq \cdots \subseteq F_k$ such that

$$\|(c_{j_\ell} - c) \restriction F_{\ell-1}\|_1 < \eta \quad \text{and} \quad \|c_{j_\ell} \restriction (A \setminus F_\ell)\|_1 < \eta;$$

the first condition holds for all large indices by pointwise convergence, and the second for a large enough finite set. Define $\lambda \in \{0,1\}^A$ by $\lambda(a) = 1$ exactly when $a \in F$ and $c(a) > 0$, or $a \in F_\ell \setminus F_{\ell-1}$ and $c_{j_\ell}(a) > 0$ for some $\ell \le k$.

Let $\ell \le k$ be given. The deficit of λ for h_{j_ℓ} is

$$\sup h_{j_\ell} - h_{j_\ell}(\delta_\lambda) = \sum_{\lambda(a)=0} c_{j_\ell}(a)^+ + \sum_{\lambda(a)=1} c_{j_\ell}(a)^-,$$

and we bound its contribution on four disjoint pieces of A. On $F_\ell \setminus F_{\ell-1}$ the sign pattern of λ is that of c_{j_ℓ}, and the contribution is 0. On F it is that of c, and a label where the signs of c_{j_ℓ} and c disagree contributes at most $|c_{j_\ell}(a) - c(a)|$; the total is less than η. On $F_{\ell-1} \setminus F$ the contribution is at most $\|c_{j_\ell} \restriction (F_{\ell-1} \setminus F)\|_1 \le \|(c_{j_\ell} - c) \restriction F_{\ell-1}\|_1 + \|c \restriction (A \setminus F)\|_1 < 2\eta$. On $A \setminus F_\ell$ it is less than η. Hence $h_{j_\ell}(\delta_\lambda) > \sup h_{j_\ell} - 4\eta \ge \beta - 4\eta > \alpha$. □

Corollary 8.2. *For the Rado observable, no tree property of a bounded affine combination of responses is witnessed by actual preparations, although the real-valued theory of $\mathcal{S}_E$ has the tree property of the second kind.*

Proof. By Lemma 5.5, the Rado observable has freely independent pure responses, and Theorem 8.1 applies. The second statement is Theorem 6.1. □

The finite arrays in the proof of Theorem 6.1 are made of actual preparations, and their rows have any prescribed finite length; by Theorem 8.1, the rows cannot be made infinite in $\mathcal{S}_E$. The infinite rows obtained by compactness in an $\aleph_1$-saturated elementary extension must therefore contain idealized preparations. The same holds for the random bipartite graph, whose observable has freely independent pure responses by its extension property.

8.2. **Quantifier-free formulas.** Theorem 8.1 concerns affine combinations of responses. In the real-valued logic of the second paper, a quantifier-free formula in a variable x of the sort of probes, built from the response with parameters of the sort of preparations, is a continuous function of finitely many responses, since the responses take values in $[0,1]$ and the connectives are continuous functions. The next theorem is the analogue of Theorem 8.1 for these formulas. Its proof uses the same gliding hump, with one change: a continuous function of responses is no longer maximized by reading off signs (we show this after the proof), so instead of building the probe from signs we glue maximizers of the individual instances on disjoint blocks of labels.

Theorem 8.3 (Compatibility of quantifier-free conditions). *Let C_φ be a diagonal observable with freely independent pure responses, let $n \geq 1$ be an integer, and let $u\colon [0,1]^n \to \mathbb{R}$ be a continuous function. Let $\mathcal{R}_u$ be the family of functions*

$$h(\nu) = u\big(r_{\mu_1}(\nu), \dots, r_{\mu_n}(\nu)\big) \qquad (\nu \in \mathcal{P}(\{0,1\}^A)),$$

where $\mu_1, \dots, \mu_n$ range over the mixtures of Alice. Then the family $\mathcal{R}_u$ contains no uniformly k-inconsistent sequence, for any k. In particular, no tree property of the quantifier-free formula $u(R(y_1,x), \dots, R(y_n,x))$ in the probe variable x is witnessed by actual preparations.

Proof. For $\nu \in \mathcal{P}(\{0,1\}^A)$ write $\pi(\nu) \in [0,1]^A$ for its vector of marginals $(\pi_a(\nu))_{a\in A}$, defined before Theorem 8.1. Every $h \in \mathcal{R}_u$ factors as $h(\nu) = H(\pi(\nu))$ with

$$H(q) = u\big(\langle \mu_1, q\rangle, \dots, \langle \mu_n, q\rangle\big), \qquad \langle \mu, q\rangle = \sum_{a\in A} \mu(a) q_a,$$

and every $q \in [0,1]^A$ is the vector of marginals of a product measure. It therefore suffices to work on the cube $[0,1]^A$.

Let (h_j) be a sequence in $\mathcal{R}_u$, with parameters $\mu_{j,1}, \dots, \mu_{j,n}$, and let $\alpha < \beta$ be such that each h_j takes a value $\geq \beta$. We show that for every k there are $j_1 < \cdots < j_k$ and $q \in [0,1]^A$ with $H_{j_\ell}(q) > \alpha$ for all ℓ. Since u is uniformly continuous, there is $\theta > 0$ such that $|u(s) - u(t)| < \beta - \alpha$ whenever $\max_i |s_i - t_i| < \theta$. For each j choose $q^{(j)} \in [0,1]^A$ with $H_j(q^{(j)}) \geq \beta$. The set A is countable and the cube is compact and metrizable, so a diagonal argument provides a subsequence, still indexed by j, along which $\mu_{j,i} \to \mu_i$ pointwise on A for each $i \leq n$ and $q^{(j)} \to q^\infty$ in $[0,1]^A$. By Fatou's lemma, each limit μ_i has total mass at most 1. Put $\eta = \theta/5$ and choose a finite set $F \subseteq A$ with $\mu_i(A \setminus F) < \eta$ for all $i \leq n$.

Put $F_0 = F$, and choose inductively indices $j_1 < \cdots < j_k$ and finite sets $F_0 \subseteq F_1 \subseteq \cdots \subseteq F_k$ such that, for all $i \leq n$,

$$\sum_{a\in F_{\ell-1}} |\mu_{j_\ell,i}(a) - \mu_i(a)| < \eta, \qquad \sum_{a\in F} |q_a^{(j_\ell)} - q_a^\infty| < \eta, \qquad \mu_{j_\ell,i}(A \setminus F_\ell) < \eta.$$

The first two conditions hold for all large indices, by convergence on the finite sets $F_{\ell-1}$ and F, and the third for a large enough finite set F_ℓ. Define $q \in [0,1]^A$ by $q_a = q_a^\infty$ for $a \in F$, $q_a = q_a^{(j_\ell)}$ for $a \in F_\ell \setminus F_{\ell-1}$ and $\ell \leq k$, and $q_a = 0$ for $a \notin F_k$.

Let $\ell \leq k$ and $i \leq n$ be given. We compare $\langle \mu_{j_\ell,i}, q\rangle$ with $\langle \mu_{j_\ell,i}, q^{(j_\ell)}\rangle$, on the same four pieces of A as in the proof of Theorem 8.1; on each piece the two vectors differ by at most 1 in each coordinate. On $F_\ell \setminus F_{\ell-1}$ they agree. On F the difference is at most $\sum_{a\in F} |q_a^\infty - q_a^{(j_\ell)}| < \eta$, since $\mu_{j_\ell,i}(a) \leq 1$. On $F_{\ell-1} \setminus F$ it is at most $\mu_{j_\ell,i}(F_{\ell-1} \setminus F) \leq \sum_{a\in F_{\ell-1}} |\mu_{j_\ell,i}(a) - \mu_i(a)| + \mu_i(A \setminus$

$F) < 2\eta$. On $A \setminus F_\ell$ it is at most $\mu_{j_\ell,i}(A \setminus F_\ell) < \eta$. Hence $|\langle \mu_{j_\ell,i}, q\rangle - \langle \mu_{j_\ell,i}, q^{(j_\ell)}\rangle| < 4\eta < \theta$ for every $i \le n$, and the choice of θ gives $H_{j_\ell}(q) > H_{j_\ell}(q^{(j_\ell)}) - (\beta - \alpha) \ge \alpha$. □

For affine combinations, the proof of Theorem 8.1 yields slightly more: there the common witness can be taken pure. For an arbitrary connective u the witness is the product probe whose vector of marginals is q, and it cannot always be taken pure: for $n = 1$, $\mu = \delta_a$ and $u(t) = 1 - |2t - 1|$, the instance vanishes at every pure probe and equals 1 at the probes with $q_a = 1/2$.

The difference between the two paradigms lies entirely in the probe space: on $\{0,1\}^A$, affine combinations of coordinates see only marginals, and conditions on marginals of disjoint blocks are compatible. On the group V^*, the linear relations among the labels make every product of coordinates affine in the coordinates, and they put the indicators of syndrome sectors in the affine span of the responses (see Proposition 7.1).

9. Affine logic and correlated preparations

Finally, we place the results in affine logic, in the form developed by Ben Yaacov, Ibarlucía and Tsankov [BYIT24], whose connectives are the affine maps and in which a mixture of structures is a direct integral. Formula values in a direct integral are integrals of formula values, which is the rule by which a response is computed on a mixed preparation. In this logic, a structure is *extremal* when it realizes only extreme types. Ben Yaacov, Ibarlucía and Tsankov [BYIT24] showed that every model of an affine theory is an affine elementary substructure of a direct integral of extremal models, and that every model of a simplicial theory, one whose type spaces are Choquet simplices, is such a direct integral, uniquely.

Ben Yaacov and Ibarlucía [BYI26] proved that an affine formula stable in almost every member of a measurable field of structures is stable in its direct integral, and that an affine theory is stable exactly when its extremal models are stable. Their definition of stability is the double limit condition used in the first paper of this series [DI26a]. Mixing in the sense of this paper behaves differently with respect to simplicity: the pure structures of the Pauli group and of the Rado graph are simple, whereas, by Theorem 6.1, the structures of their observables have the tree property of the second kind.

The direct integral allows the elements in the two variables of a formula to depend on the same random parameter. For observables this means that Alice and Bob share randomness: they prepare a separable state, a mixture of product states, rather than a single product state. Keisler's randomizations [Kei99, BY13] are of this kind, while Theorem 6.1 concerns product preparations.

10. Questions

Question 10.1. Which diagonal observables with the independence property have the tree property of the second kind witnessed by actual preparations? Theorems 7.2 and 8.1 and Proposition 7.1 suggest a criterion in terms of the linear span of the responses in $C(K_\varphi)$: whether it contains, with coefficients of bounded total absolute value, the indicators of an infinite independent sequence of nested clopen sets.

Question 10.2. By Remark 7.3, the proof of Theorem 7.2 adapts to qudits of prime dimension. For continuous variables, the commutation phase of a translation by x and a momentum kick by k is the interference fringe e^{ikx}, which is not NIP by the second paper. Where does the unbounded fringe lie with respect to simplicity?

Question 10.3. In classical model theory, a simple theory without the independence property is stable. Is a simple observable stable? By Theorem 6.1, a simple diagonal observable has no independence property, so the question is whether the unstable observables of the second paper, such as the comparison of photon numbers, have a tree property once preparations are mixed.

Question 10.4. Does Theorem 6.1 hold for every observable whose responses have the independence property, not only for diagonal ones? The proof uses the values 0 and 1 of pure responses; for general observables they are replaced by two distinct levels, and the averages must be controlled with a gap.

Question 10.5. Is there a theorem of Kim and Pillay for affine logic, characterizing simple affine theories by an independence relation on extremal types?

Historical remarks

Simple theories were introduced by Shelah [She80] in 1980, together with the tree properties of the first and second kind; the motivating example was T_{ind}, the theory of the countable random graph (see [She80, GIL02]). Kim [Kim98] proved in 1998 that forking is symmetric in simple theories, and Kim and Pillay [KP97] proved the independence theorem and the characterization of simplicity by an independence relation in 1997. Ben Yaacov [BY03] extended the theory to compact abstract theories and to continuous logic. The Rado graph was described by Ackermann [Ack37] in 1937 and by Rado [Rad64] in 1964; by a theorem of Erdős and Rényi [ER63] from 1963, it is the almost sure isomorphism type of the random graph on countably many vertices. Fraïssé [Fra54] introduced his limits in 1954, and the theorem of Witt on isometries of forms is in Artin's *Geometric Algebra* [Art57]. Infinite-dimensional vector spaces over finite fields with a nondegenerate form are among the smoothly approximable structures of Cherlin and Hrushovski [CH03]. Granger [Gra99] studied vector spaces with a generic bilinear form over an arbitrary field in his thesis of 1999 and showed that, over an infinite field carried as a second sort, the theory is not simple; Chernikov and Ramsey [CR16] proved in 2016 that over an algebraically closed field it is NSOP_1. The symplectic description of the Pauli group and of stabilizer codes is due to Gottesman [Got97] and to Calderbank, Rains, Shor and Sloane [CRSS98]; the expression of the projector onto a syndrome sector as an average over the stabilizer group, behind Proposition 7.1, is standard in the stabilizer formalism of Gottesman. Keisler [Kei99] introduced randomizations in 1999; Ben Yaacov and Keisler [BYK09] treated them as metric structures, and Ben Yaacov [BY13] proved in 2013 that the randomization of a theory with the independence property has TP_2. Direct integrals and extremal models in affine logic are due to Ben Yaacov, Ibarlucía and Tsankov [BYIT24], and stability in affine logic to Ben Yaacov and Ibarlucía [BYI26]. The first two papers of this series are [DI26a, DI26b].

Declarations

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